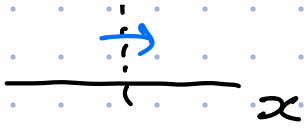
DIFFUSION

RATE [PARTICLES PER SECOND]

$\lambda = J$ IN 1d

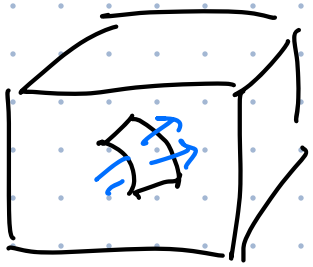


PS3

$\lambda = \int J \, dx$ IN 2d

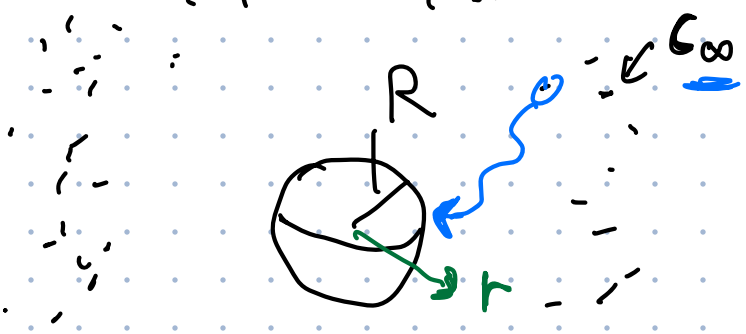


$\lambda = \iint J \, dA$ IN 3d



EXAMPLE

FIND THE ARRIVAL RATE (PARTICLES PER SECOND) ARRIVING AT A PERFECTLY ABSORBING SPHERE IN A UNIFORM CONCENTRATION FAR AWAY



FAR FIELD CONCENTRATION

$c(r) \to C_\infty$

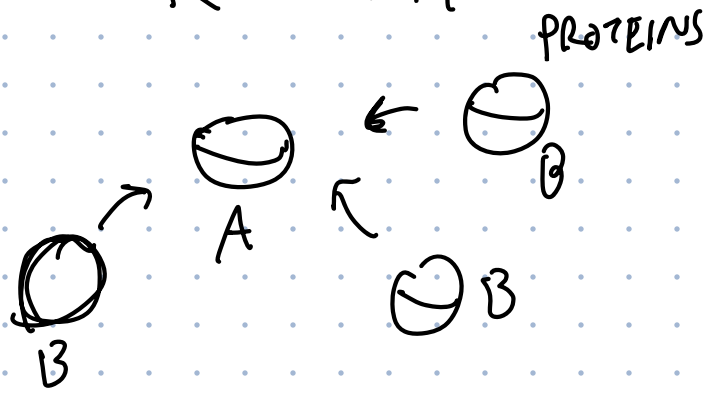
AS $r \to \infty$

PERFECT ABSORPTION

$c(R) = 0$

CASE I

$R \sim 5\,nM$

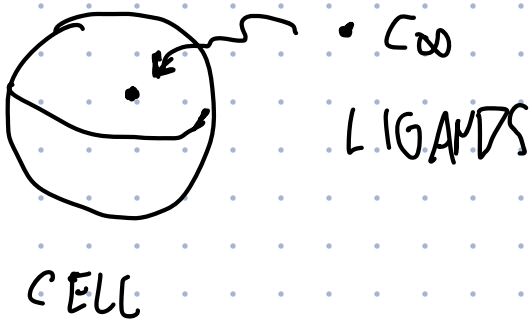


$A + B \xrightarrow{\lambda} C$

$A + B \xrightarrow{\lambda} A + B^{*}$

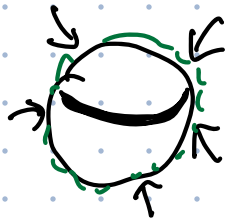
CASE II

$R \sim 10\,\mu m$



SOLVE: BY SYMMETRY $c = c(r)$

- $c(R) = 0$ B.C.
- $c(r) \to c_\infty$ AS $r \to \infty$



WE SEEK $\lambda = -\iint J \cdot dA$

$= -J \cdot A$

$= -4\pi R^2 J$

AT STEADY STATE,

$$\frac{\partial c}{\partial t} = D \nabla^2 c$$

$$0 = D \nabla^2 c$$

$$0 = D \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c}{\partial r} \right) \right) \quad \star$$

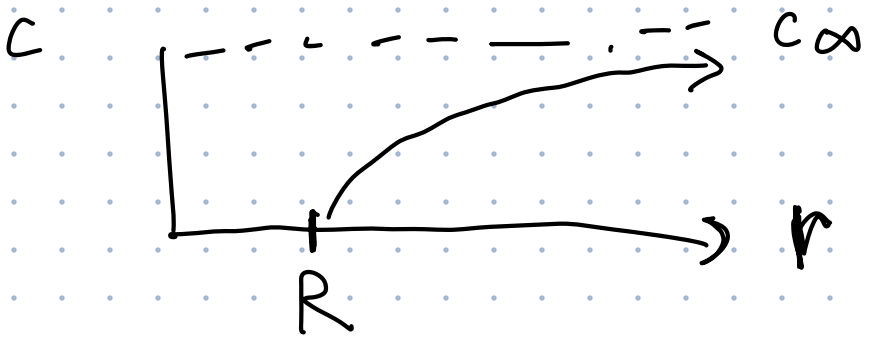
$$\cdots \quad 0 = \frac{\partial}{\partial r} \left(r^2 \frac{\partial c}{\partial r} \right)$$

$$r^2 \frac{\partial c}{\partial r} = A \Rightarrow \frac{\partial c}{\partial r} = \frac{A}{r^2}$$

$$c(r) = -\frac{A}{r} + B$$

PLUG IN B.C

$$c(r) = c_\infty \left(1 - \frac{R}{r} \right)$$



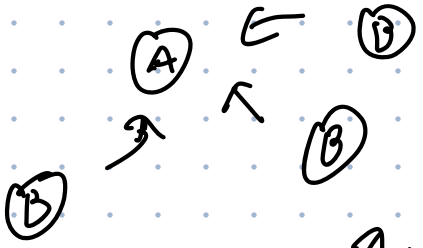
$$\vec{J}(r) = -D\nabla c$$

$$= -D\frac{\partial c}{\partial r}$$

$$J(R) \approx -D\frac{c_\infty}{R}$$

$$\boxed{\lambda = 4\pi R D c_\infty} \quad \star$$

CASE I:



$$A + B \xrightarrow{\lambda} m$$

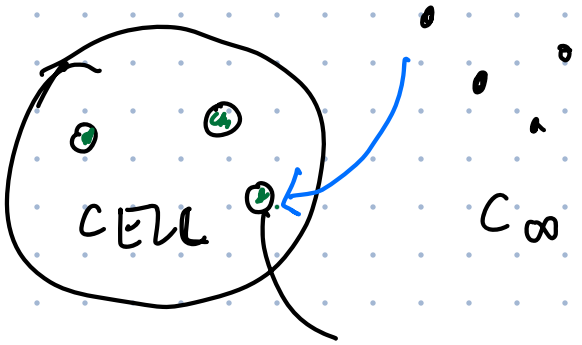
$R \sim 5\,nM$

$D \sim 10\,\mu m^2/s$

$\lambda \sim (100\,\mu M^{-1} s^{-1}) c_\infty$

DIFFUSION-LIMITED RATE

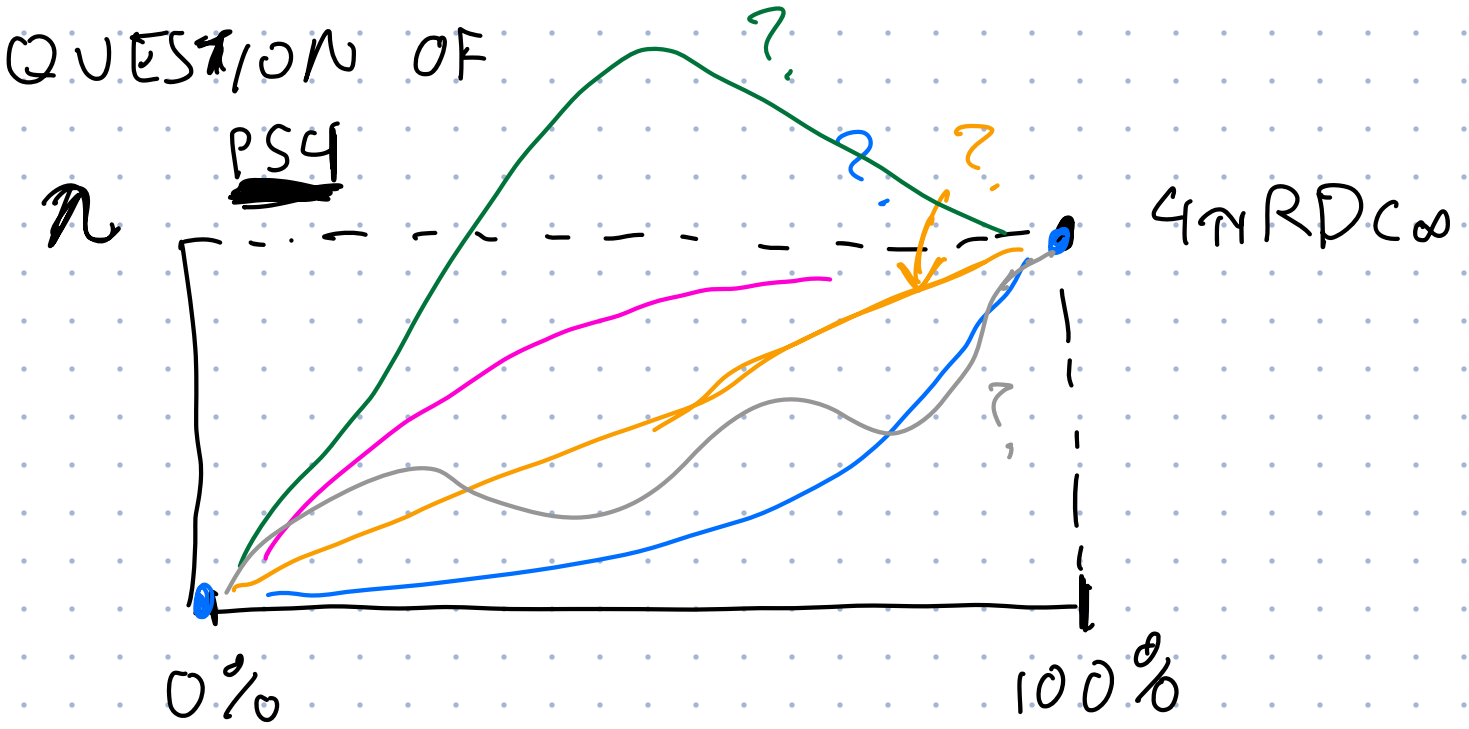
CASE II



$R \sim 10\,\mu m$

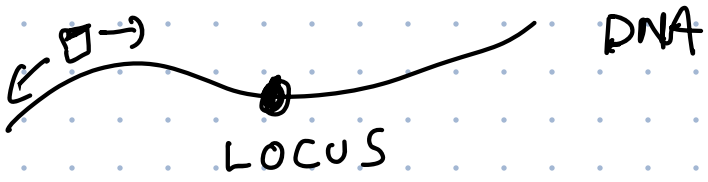
$\lambda = 4\pi R D c_\infty$

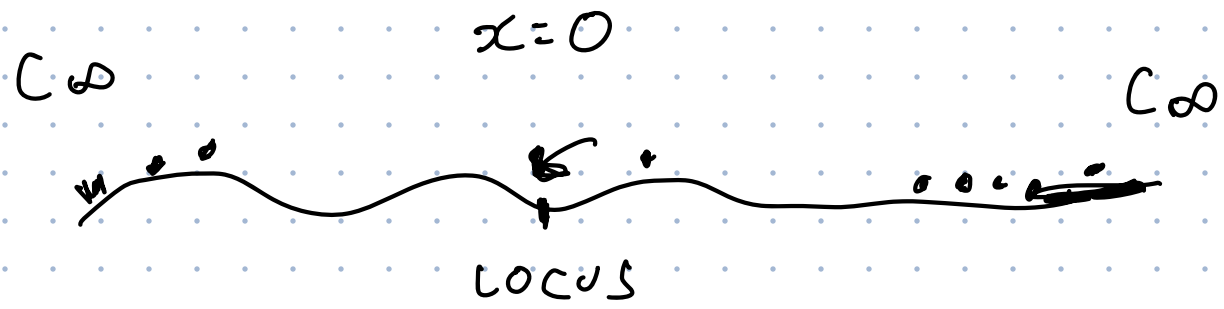
ASSUMING THE CELL IS 100% COVERED IN RECEPTORS.



IN 1d

EXAMPLE:





PERFECT ABSORPTION $c(0) = 0$ ★

FAR FIELD $c(x) \to c_\infty$ ★
AS $x \to \pm\infty$

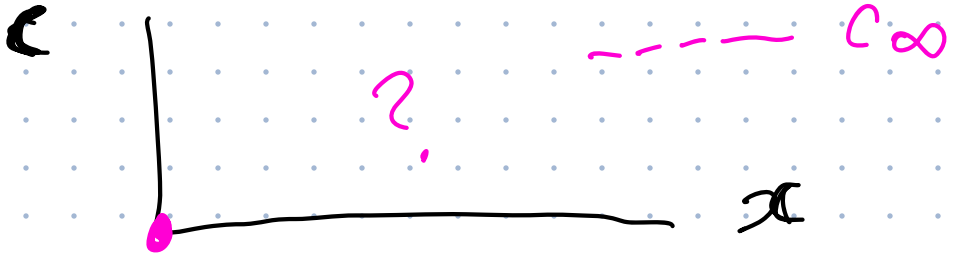
STEADY STATE

$$0 = D \frac{\partial^2 c}{\partial x^2}$$ ★

$$0 = \frac{\partial^2 c}{\partial x^2} \Rightarrow \frac{\partial c}{\partial x} = A$$

$$c(x) = Ax + B$$

NO SOLUTION



EXAMPLE

